Notes on the Fourier transform

Convention: time t in seconds; frequency ν in Hz.
Illustrative companion sheet · Design preview

The Fourier transform

$$\hat{f}(\nu) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \nu t}\, dt$$

The inverse transform uses the positive exponential.

For the example signal

f(t) = cos(2π · 3t) + 0.45 cos(2π · 7t),

the two positive-frequency components are at 3 and 7 Hz.
The figure uses a one-sided amplitude spectrum.

A finite sampling interval produces spectral leakage
when components do not fall on discrete frequency bins.

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